(1) (a) $\sum_{n=-\infty}^{\infty} t^n J_n(x+y) = \exp\left[\frac{(x+y)}{2}\left(t-\frac{1}{t}\right)\right]$

(1) $= e^{\frac{x}{2}\left(t-\frac{1}{t}\right)} \cdot e^{\frac{y}{2}\left(t-\frac{1}{t}\right)}$

$$= \left(\sum_{m=-\infty}^{\infty} t^m J_m(x)\right)\left(\sum_{k=-\infty}^{\infty} t^k J_k(y)\right)$$

$$= \sum_m \sum_k t^{m+k} J_m(x) J_k(y).$$

Let $z = m+k$.

$$\therefore \sum_z \sum_k t^z \cdot J_{z-k}(x) J_k(y) = \sum_z \left(\sum_k J_{z-k}(x) J_k(y)\right) t^z$$

Comparing with (1):

$$\boxed{J_n(x+y) = \sum_{m=-\infty}^{\infty} J_{n-m}(x) J_m(y)}$$ [Renaming $k \to m$]

(Proved)

(b) $e^{\frac{x}{2}\left(t-\frac{1}{t}\right)} = \sum_{n=-\infty}^{\infty} t^n J_n(x)$

Put $t = ie^{i\theta}$.

$$\therefore \frac{x}{2}\left(t-\frac{1}{t}\right) = \frac{x}{2}\left[ie^{i\theta} + ie^{-i\theta}\right]$$
$$= ix\cos\theta$$

$\therefore$ LHS : $e^{ix\cos\theta}$

RHS : $$\boxed{\sum_{n=-\infty}^{\infty} i^n e^{in\theta} J_n(x) = e^{ix\cos\theta}}$$ (Proved)

(c) $J_0(x) = \sum_{n=0}^{\infty} (-1)^n \frac{1}{(n!)^2} \left(\frac{x}{2}\right)^{2n}$

$$\therefore J_0(x\cos\theta) = \sum_{n=0}^{\infty} \frac{(-1)^n}{(n!)^2 2^{2n}} (x\cos\theta)^{2n} \cdot \cos\theta$$

$\cos\theta$

$$= \sum_{n=0}^{\infty} \frac{(-1)^n}{(n!)^2 2^{2n}} x^{2n} (\cos\theta)^{2n+1}$$

$$\int_0^{\pi/2} J_0(x\cos\theta)\cos\theta \, d\theta$$

$$= \sum_{n=0}^{\infty} \frac{(-1)^n}{(n!)^2} \frac{x^{2n}}{2^{2n}} \int_0^{\pi/2} (\cos\theta)^{2n+1} \, d\theta$$

$$= \sum \left[\frac{(-1)^n x^{2n}}{(n!)^2 \, 2^{2n}} \right] \cdot \frac{1}{2} \frac{\sqrt{\pi}\, n!}{(n+\frac{1}{2})(n-\frac{1}{2}) \cdots \frac{1}{2} \cdot \sqrt{\pi}}$$

$$= \frac{(-1)^n x^{2n}}{n! \, 2^{2n}} \cdot \frac{1}{2} \cdot \frac{2^{n+1}}{(2n+1)(2n-1) \cdots 1}$$

$$= \frac{(-1)^n x^{2n}}{n! \, 2^{2n}} \cdot \frac{2^n (2n)(2n-2) \cdots 2}{(2n+1)!}$$

$$= \frac{(-1)^n x^{2n}}{n! \, 2^n} \cdot \frac{2^n n!}{(2n+1)!} = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n+1)!}$$

$$= \frac{1}{x} \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!}$$

$$\boxed{\int_0^{\pi/2} J_0(x\cos\theta)\cos\theta \, d\theta = \frac{\sin x}{x}}$$ (Proved)

(3) $$e^{\frac{x}{2}(t-\frac{1}{t})} = \sum_{n=-\infty}^{\infty} t^n J_n(x). \quad \text{— (1)}$$

$\therefore$ Differentiating both sides wrt x:

$$\frac{1}{2}\left(t-\frac{1}{t}\right) e^{\frac{x}{2}(t-\frac{1}{t})} = \sum_{n=-\infty}^{\infty} t^n J_n'(x)$$

$$\Rightarrow \left(t-\frac{1}{t}\right) e^{\frac{x}{2}(t-\frac{1}{t})} = 2\sum_n t^n J_n'(x).$$

$$\Rightarrow \sum_n t^{n+1} J_n(x) - \sum_n t^{n-1} J_n(x) = 2\sum_n t^n J_n'(x) \quad \text{[from (1)]}$$

Comparing the coefficient of t^n:

$$\boxed{J_{n-1} - J_{n+1} = 2J_n'(x)}$$ (Proved)

(4) $$j_n(x) = \sqrt{\frac{\pi}{2x}} J_{n+\frac{1}{2}}(x)$$

$$j_1 = \sqrt{\frac{\pi}{2x}} J_{\frac{3}{2}} = \sqrt{\frac{\pi}{2x}} \sum_{n=0}^{\infty} \frac{(-1)^n}{n!\,\Gamma(n+\frac{3}{2}+1)} \cdot \left(\frac{x}{2}\right)^{2n+\frac{3}{2}}$$

$$\Gamma\left(n+\frac{3}{2}+1\right) = \left(n+\frac{3}{2}\right)\Gamma\left(n+\frac{3}{2}\right)$$

$$= \left(n+\frac{3}{2}\right)\left(n+\frac{1}{2}\right)\Gamma\left(n+\frac{1}{2}\right)$$

$$= \left(n+\frac{3}{2}\right)\left(n+\frac{1}{2}\right)\cdots\cdots\frac{1}{2}\Gamma(1/2)$$

$$= \frac{1}{2^{n+2}}(2n+3)(2n+1)\cdots\cdots 1\sqrt{\pi}$$

$$= \frac{(2n+3)!}{2^{n+2}(2n+2)(2n)\cdots\cdots 2}\sqrt{\pi}$$

$$\boxed{\Gamma\left(n+\frac{3}{2}+1\right) = \frac{(2n+3)!}{2^{2n+3}(n+1)!}\sqrt{\pi}}$$

$$\therefore J_1 = \cancel{\sqrt{\frac{\pi}{2}}}\sum_{n=0}^{\infty}\frac{(-1)^n(n+1)!\,2^{2n+3}}{n!\,(2n+3)!\,\cancel{\sqrt{\pi}}}\cdot\frac{x^{2n+1}}{2^{2n+2}}$$

$$= 2\sum_{n=0}^{\infty}\frac{(-1)^n(n+1)}{n!\,(2n+3)!}x^{2n+1}$$

$$\therefore -\frac{\cos x}{x} + \frac{\sin x}{x^2} = -\sum_{n=0}^{\infty}\frac{x^{2n-1}}{(2n)!}(-1)^n + \sum_{n=0}^{\infty}\frac{x^{2n-1}}{(2n+1)!}(-1)^n$$

$$= \sum\frac{x^{2n-1}}{(2n)!}\left[\sum_{n=0}^{\infty}(-1)^n\left[\frac{2}{(2n+1)}-1\right]\right]$$

$$= \sum\frac{-x^{2n-1}}{(2n)!}\left[\sum_{n=0}^{\infty}\frac{(-1)^n\,2n}{(2n+1)}\right]$$

$$= -\sum_{n=0}^{\infty}\frac{(-1)^{n+1}\cdot 2n\,x^{2n-1}}{(2n+1)!}$$

$n \to n+1$.

$$-\frac{\cos x}{x} + \frac{\sin x}{x^2} = 2\sum_{n=0}^{\infty}\frac{(-1)^n(n+1)\,x^{2n+1}}{(2n+3)!}$$ [$n=-1$ term is 0]

$$\Rightarrow \boxed{-\frac{\cos x}{x} + \frac{\sin x}{x^2} = J_1(x)}$$ (Proved)

7. $$f(x) = \begin{cases} 1 & ; \ 0 < x \le 1 \\ -1 & ; \ -1 \le x < 0 \end{cases}$$

Let $f(x) = \sum_{l=0}^{\infty} a_l P_l(x)$

$$\Rightarrow \int_{-1}^{1} f(x)P_n(x) = \sum_{l=0}^{\infty} a_l \int_{-1}^{1} P_l P_n$$

$$= \sum_{l=0}^{\infty} a_l \frac{2}{2n+1}\delta_{ln} = \frac{2a_n}{2n+1}.$$

$$\Rightarrow \quad 2m \frac{(2n+1)}{2} \int_{-1}^{1} f(x)\, P_n(x)\, dx = a_n$$

$$\Rightarrow \quad a_n \cancel{= \frac{(2n+1)}{2} \left[-\int_{-1}^{0} P_n(x)\, dx + \int_{0}^{1} P_n(x)\, dx \right]}$$

$f(x) = -f(-x) \longrightarrow$ odd f^n.

$\therefore$ $f(x)\, P_n(x) \longrightarrow$ odd f^n if n = odd even [even × odd]

$\longrightarrow$ even " " n = even odd [odd × odd]

$\therefore$ If n = odd even ; the integral vanishes.

If n = ~~even~~ odd :

$$\int_{-1}^{1} f(x)\, P_n(x)\, dx = 2\int_{0}^{1} f(x)\, P_n(x)\, dx = 2\int_{0}^{1} P_n(x)\, dx$$

$$\therefore \quad a_n = (2n+1) \int_{0}^{1} P_n(x)\, dx.$$

$$(2n+1)\, P_n = P'_{n+1} - P'_{n-1}.$$

$$\Rightarrow \quad 2n+1 \int_{0}^{1} P_n(x)\, dx = -P_{n-1}\Big|_0^1 + P_{n+1}\Big|_0^1$$

$$= -P_{n-1}(1) + P_{n-1}(0) + P_{n+1}(1) - P_{n+1}(0)$$

$$= -P_{n+1}(0) + P_{n-1}(0)$$

[$P_n(1) = 1$]

$$\therefore \quad a_n = -P_{n+1}(0) + P_{n-1}(0)$$

$$\therefore \quad \boxed{f(x) = \sum_{k=0}^{\infty} \left[-P_{2k+2}(0) + P_{2k}(0) \right] P_{2k+1}(x)}$$

[only terms appear : $n \to 2k+1$]

(Ans)